

SMART INDIA HACKATHON 2026

Honey Chain

Complete technology stack: hardware, AI and software

Problem Statement **SIH 26021**, Ministry of MSME (KVIC)

Theme: Agriculture, FoodTech & Rural Development

Team: **Say My Name**

AI assists. KVIC authorises. Blockchain records. QR verifies.

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|--|--------------------------------------|
| 1. Smart hive hardware (IoT) | 6. Weather and alerts (software GSM) |
| 2. Edge processing and the hive that listens | 7. Blockchain and traceability |
| 3. IoT gateway | 8. Software platform |
| 4. AI / ML models and datasets | 9. One-slide summary |
| 5. AI voice assistant: Madhu Mitra | |

BUILT

working in the prototype

PARTLY

part working, upgrade planned

PLANNED

next phase

1 Smart hive hardware (IoT)

Component	Purpose	Status
ESP32 microcontroller	Reads all sensors, processes sound, sends data over Wi-Fi	BUILT
DHT11 ×2 → DS18B20 / SHT31	Brood-nest temperature and humidity (upgrade to ±0.5 °C probes)	PARTLY
MQ gas + CO ₂ sensor → SCD40 / MH-Z19B	CO ₂ in ppm and air quality inside the hive	PARTLY
Load cell + HX711	Hive weight: nectar flow, falling stores, swarm (sudden 1–3 kg drop)	FIRMWARE READY
BME280 (outside the hive)	Outside temperature, humidity and air pressure: inside-vs-outside comparison, storm warning	PLANNED
INMP441 I ² S MEMS microphone	Hive sound: queen status, swarm preparation	PLANNED
Relay + 12 V fan	Cooling for heat only. High CO ₂ raises an alert; the fan stays off	BUILT
Relay + heater bulb	Keeps the brood warm on cold nights	BUILT
SSD1306 OLED (0.96")	Live readings shown at the hive	BUILT
USB camera / ESP32-CAM	Entrance video: varroa mites, bee activity	PARTLY
Solar panel + Li-ion battery, LoRa	Rural sites with no mains power or Wi-Fi	PLANNED

2 Edge processing and the hive that listens

All noise cleaning is done in software on the ESP32, so the hardware stays cheap and small.

Technology	Purpose	Status
Arduino C++ firmware	Sensor loop, fan and heater control with hysteresis, local web dashboard	BUILT
I ² S digital audio bus	24-bit digital sound from the mic chip, so the wires pick up no noise	PLANNED
ESP-DSP: high-pass + low-pass filters	Removes wind and machinery below 100 Hz, birds and hiss above 600 Hz	PLANNED
Rain gate	Skips moments when rain drums on the roof (broadband noise)	PLANNED
ESP-DSP: FFT	Splits the 100–600 Hz buzz into its main frequencies (colony hum, queen piping)	PLANNED
TensorFlow Lite Micro / Edge Impulse	Queen-status model running on the chip; only small feature packets are sent, never raw audio	PLANNED

Sound pipeline: INMP441 mic → I²S → high-pass 100 Hz → low-pass 600 Hz → rain gate → FFT peaks + MFCC → AI model → "Queenless" / "Swarm risk" alert

3 IoT gateway

Technology	Purpose	Status
Python Flask	Receives hive data, runs the health check, serves live readings to the Keeper app	BUILT
OpenCV	Camera capture and streaming	BUILT
Alert worker	Checks hive health every 60 s and reports problems to the main server	BUILT

4 AI / ML models and datasets

Key colony-health signal: a healthy colony holds its brood at 34–35 °C whatever the weather. When the inside temperature starts following the outside temperature, the colony is weak or queenless.

Model	Used for	Dataset	Status
Rule-based expert system	Hive health score and risks, with plain-language advice	Beekeeping guidelines	BUILT
AI lab-report check	Flags lab values outside FSSAI limits for the officer	FSSAI honey standards	BUILT
Linear trend forecast	Honey production forecast	Keeper's own harvest records	BUILT
Copied-QR detection	Impossible travel, too many places or phones, buyer reports	Live scan data	BUILT
Isolation Forest	Unusual hive behaviour	HOBOS (Univ. of Würzburg), MSPB 2024	PLANNED
XGBoost	Healthy / Weak / Queenless / Swarm risk	MSPB (53 hives, inspection labels)	PLANNED
MFCC + CNN	Queen status from hive sound	Kaggle "Smart Bee Colony Monitor"	PLANNED
YOLOv8-nano	Varroa mite detection at the entrance	Roboflow Varroa, Bilik et al. 2021	PLANNED
MobileNetV2	Bee-health check from a frame photo	Kaggle BeelImage (5,000+ images)	PLANNED
Regression model	Predicts next hour's inside temperature, so heating or cooling starts early	HOBOS + Open-Meteo history	PLANNED

Tools: Python · scikit-learn · XGBoost · TensorFlow / Keras · Ultralytics YOLO · Jupyter

5 AI voice assistant: "Madhu Mitra"

The keeper asks any beekeeping doubt by voice, in their own language, and hears the answer.

Layer	Technology	Status
Speech → text	Bhashini ASR (MeitY) · OpenAI Whisper · AI4Bharat IndicConformer	PLANNED
AI brain	LLM: Claude / Gemini API, or Llama 3 offline via Ollama	PLANNED
Trusted knowledge (RAG)	pgvector + bge-m3 embeddings over NBB, KVIC, ICAR and FSSAI manuals	PLANNED
Live hive answers	LLM function calling into the Honey Chain API (hive status, alerts, batches, weather)	PLANNED
Text → speech	Bhashini TTS · AI4Bharat Indic-Parler-TTS	PLANNED
In the app	Web Speech API (browser) · Flutter speech_to_text + flutter_tts (Android)	PLANNED
Phone line for basic phones	Exotel / Twilio Voice IVR: missed call → the bot calls back	PLANNED
Safety	Answers cite their source; no antibiotic doses; "Talk to officer" opens a help-desk ticket	PLANNED

Flow: Keeper speaks (Tamil / Hindi / ...) → Bhashini speech-to-text → trusted manuals (RAG) + the keeper's live hive data → LLM answer → Bhashini text-to-speech → keeper hears the answer

6 Weather and alerts ("software GSM")

No GSM module or SIM card in the hive: the cloud AI decides, and the keeper is reached on their phone.

Technology	Purpose	Status
SMS gateway API (Fast2SMS / Twilio)	Critical hive alerts and OTPs by SMS, with TRAI DLT-registered templates	BUILT
In-app notifications	Every alert is also saved in the Keeper app (My Hives)	BUILT
Open-Meteo Forecast API	Storm, wind, rain and heatwave alerts for each apiary's location	PLANNED
BME280 pressure-drop rule	A fall of 3 hPa or more in 3 hours means a storm is coming, even offline	PLANNED
Firebase Cloud Messaging	Free push notifications on the phone	PLANNED
Automatic voice call (Exotel / Twilio)	A critical alert not opened in 15 min triggers a phone call in the keeper's language	PLANNED
WhatsApp Business Cloud API	Alert with a photo or chart	PLANNED

Escalation: Push notification → SMS → automatic voice call → regional KVIC officer notified if the problem continues or spreads across an area

SMS alerts run in demo mode until an SMS provider key is added. Critical (HIGH) problems are texted at once; the same problem is repeated at most every 6 hours.

7 Blockchain and traceability

Technology	Purpose	Status
Solidity smart contract	Records harvest, lab certificate, officer approval, jars and sales	BUILT
Hardhat local chain · Polygon Amoy ready	Blockchain network (local for the demo, public testnet ready)	BUILT
ethers.js · EIP-712 signatures	The keeper and the officer each sign their own step	BUILT
Merkle tree	The number of jar QR codes is capped by the honey actually harvested	BUILT
IPFS CIDv1	Content IDs for batch and lab metadata	BUILT
Hash-chained ledger + database triggers	Records cannot be edited or deleted	BUILT
QR code per jar	Consumer verification with no app or login	BUILT
Keeper OTP confirmation	A wholesaler records honey received; the keeper confirms with an OTP; logged on the ledger	BUILT

8 Software platform

Layer	Technology	Status
Keeper app · KVIC Admin portal · public website	React 19, Vite, Tailwind CSS, Recharts, Lucide icons	BUILT
Installable app	Progressive Web App (vite-plugin-pwa)	BUILT
Android apps	Flutter	BUILT
Backend API	Node.js + Express	BUILT
Database	PostgreSQL: private schema per organisation, row-level security	BUILT
Offline use	Harvests saved offline and synced when the internet returns	BUILT
Languages	Custom offline i18n engine, 12 Indian languages → Bhashini for live translation	PARTLY
Security	JWT, bcrypt, Helmet, rate limiting, hashed OTP (5 min), Aadhaar Verhoeff check (last 4 digits stored)	BUILT
Registration	NBB Madhukranti categories: Individual, Firm, Society, Company/FPO, Wholesaler, all approved by KVIC officers	BUILT
Digital twin	Unity 6 (playable in a browser via WebGL) + Blender 3D hive	BUILT

9 One-slide summary

Hardware	ESP32 · temperature, humidity, CO ₂ , weight and pressure sensors · INMP441 mic · fan and heater · solar
Edge	Arduino C++ · I ² S · ESP-DSP (filters + FFT) · TensorFlow Lite Micro
AI	Expert system · Isolation Forest · XGBoost · audio CNN · YOLOv8 · MobileNetV2 · LLM + RAG voice assistant
Voice & language	Bhashini (speech, translation) · 12 Indian languages
Alerts	SMS gateway API (no GSM module) · Firebase push · voice call · Open-Meteo weather
Blockchain	Solidity · Hardhat / Polygon · EIP-712 · Merkle tree · IPFS
Apps	React · Vite · Tailwind · PWA · Flutter
Backend	Node.js · Express · PostgreSQL · Python Flask
Simulation	Unity 6 · Blender

If the jury asks "what works today?": everything marked **BUILT** is in the live demo. Items marked **PLANNED** are the next phase, backed by the public research datasets named above.